POID-4547081947

Dhruvit | Aug 12, 2026, 02:41 PM

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Job details

LT_Power_Cables

Dhruvit | Aug 12, 2026, 02:41 PM

Job details

Job Id	Start Date	End Date	Category	Asset	Site	Description
PH2026100008	-	-	LT	-	IIM Ahmedabad	-

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Inspection details

Prepared by	Started on	Category	Sub category	Description	Inspection location
Dhruvit	Aug 12, 2026, 02:41 PM	LT	Power Cables	-	0, 0

Status

Completed

Basic Details

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Basic Details

Inspection Points: 9/9

Title : SAP Code

Answer 10567201

Title : Cable Size

Answer
3C x 400 sq.mm Al, XLPE, Armoured

Title : Vendor Name

Answer KEI Industries Ltd.

Title : Offered Qty

Answer 6 Drums / 3000 mtr

Title : PO Details

Answer
PO No. 4500145678 dated 10-May-2026

Title : Batch Nos

Answer
KEI/AL/XLPE/2606/112 to KEI/AL/XLPE/2606/119

Title : AEML Engineer & Designation

Answer

Vikram Desai, Manager (Vendor Quality)

Title : Vendor Engineer & Designation

Answer — Amit Verma, Quality Engineer

Title : Date of Inspection

Answer — Aug 12, 2026

Characteristics _ Parameter

Page complete: 100%

Physical Checks

Inspection Points: 15/15

Title : Overall Cable Dia

Input — 48.5 mm (spec 47.0 – 50.0 mm)

Note : Diameter check with digital Vernier – conforms to approved construction.

Title : Ovality of Cable

Input — 2.0% – at limit, accepted

Title : Outer Sheath Thickness

Input — Avg 2.65 mm, Min 2.50 mm – OK

Title : Water Swellable Tape thickness

Input

0.25 mm both longitudinal & overlapping – OK

Note : Tape thickness matches BOM; overlap verified.

Title : Armour Strand Size

Input — 2.5 mm GI round wire – OK

Title : No of Armour Strands

Input — 56 Nos

Title : Diameter over Armour

Input	47.2 mm

Title : Diameter over Inner Sheath

Input	42.0 mm

Title : Thickness of Inner Sheath

Input	1.25 mm – OK

Title : Diameter over laid up

Input	39.0 mm

Title : Insulation Thickness

Input
Avg 1.80 mm, Min 1.60 mm (spec min 1.60 mm)

Title : No of Strands in conductor

Input	61 strands of 2.52 mm dia

Title : shape of conductor

Input	Compacted circular – OK

Title : Filler Diameter

Input	4.0 mm – OK

Title : Verification of Water Blocking construction in conductor

Input
Verified – matches QAP / type test construction

Acceptance Tests

Inspection Points: 15/15

Title : Conductor Resistance Tests

Input
0.118 Ω/km @ 20°C – above max; conductor resistance fail

Note : Conductor resistance fail – investigate strand / alloy.

Title : High Voltage test @3kV for 5 minutes in all cores

Input

Passed as per QAP

Note : HV kit cal valid; test witnessed – Pass.

Title : IR Test

Input

IR phase–earth 80 MΩ – below acceptance; retest pending

Title : Hot Set Test on Insulation

Input

Elongation under load 45%, permanent set 8% – Pass

Title : Tensile Strength and Elongation for Insulation & Sheath

Input

Values within IS limits before & after ageing – Pass

Title : Water Swellable Tape Swell height

Input

Swell test OK – Pass

Title : Armour Tests - Mass of Zinc Coating & Uniformity of Zinc coating

Input

Zn coating mass & uniformity within IS 3975 – Pass

Title : Tests on HDPE Outer Sheath -1. Density 2.Thickness 3. Shrinkage 4. Pressure test at high temperature 5. Tensile Strength Without ageing & After ageing 6. Elongation at break 7. Without ageing & After ageing

Input

All listed HDPE sheath tests satisfactory – Pass

Note : Density / shrinkage / pressure / TS–E – Pass.

Title : Water Penetration Test

Input

Longitudinal WP test satisfactory – Pass

Note : WP test as per QAP – satisfactory.

Title : Anti Termite & Anti Rodent check

Input
Additive present as declared; marking OK – Pass

Title : FRLS Outer Sheath -Flammability Test

Input
Flame propagated beyond mark – flammability fail

Title : FRLS Outer Sheath-Oxygen Index Test

Input OI 31.5% – Pass

Title : FRLS Outer Sheath -Temperature Index

Input Pass

Title : FRLS Outersheath -Halogen Gas Acid Evaluation

Input
Halogen acid above limit – fail

Note : HCl evaluation does not meet FRLS criteria.

Title : FRLS Outer Sheath-Smoke Density

Input
Max smoke density < specified – Pass

Note : FRLS smoke density OK per lab sheet.

Cable Rewinding on Drum

Inspection Points: 7/7

Title : Cable Appearances / Damages check

Input
Minor scuff mark cleaned; no deep cut – OK

Note : Surface finish satisfactory over offered length.

Title : Outer sheath colour

Input Black outer sheath – OK

Title : Printing & Embossing

Input

Printing clear: size, voltage grade, IS mark, manufacturer, year – OK

Title : Sequential Length Marking

Input
Verified start & end metre marks – OK

Title : MS Plates & Spinndle

Input
MS plates painted / marked – OK

Title : Weight of Cable

Input 2100 kg gross – OK

Title : Drum Length Verification

Input 250 mtr – verified

Packing & Marking

Inspection Points: 2/2

Title : Cable End Sealing

Input Proper end caps fitted – OK

Title : Drum Stencilling & Marking

Input Drum no. & arrow marking OK

Inprocess Record Verification

Inspection Points: 6/6

Title : Wire Drawing

Input
Wire drawing QC sheets reviewed – OK

Title : Core Insulation

Input Ok

Note : Thickness & spark test logs complete.

Title : Core Laying

Input
Core laying & filler arrangement as per drawing – OK

Title : Inner Sheathing

Input
Thickness logged – within limit – OK

Title : Armouring

Input Verified – OK

Title : Outersheathing

Input
Outer sheath extrusion records OK

Raw Material / Bought out Items TC Verification

Inspection Points: 5/5

Title : Al Wire Rod

Input Ok

Title : PVC Compound

Input Ok

Title : HDPE Compound

Input Verified – OK

Title : Armour GI

Input Sub-vendor TC reviewed – OK

Title : Water Swellable Tape / Yard

Input
TC verified – swell properties OK

Other Compliances for Manufacturing Facility

Inspection Points: 5/5

Title : Safety Aspects in Facility

Input Ok

Note : PPE & emergency preparedness observed OK.

Title : House Keeping

Input Ok

Title : Eco-friendly Waste Segregation, Collection and Storage Practices w.r.t Hazardous / Non-Hazardous Materials

Input Ok

Title : Calibration of Testing Instruments

Input Ok

Title : SA 8000 Compliance

Input Ok

Note : No adverse social-compliance observation.

Inspection Observations

Page complete: 100%

Default

Inspection Points: 2/2

Observations : 1.

Answer Minor

Observations : 2

Answer Minor

Note : No blocking issue on observation point 2.

Inspection Conclusion

Page complete: 100%

Inspection Summary

Inspection Points: 3/3

Title : Test Results

Answer

Not Ok

Title : Technical Clearance

Answer

Subject to rectification of minor d efects

Note : Clearance withheld – re-inspection / rejection.

Title : Declaration

Answer

Re-inspection will be required after rectification of non-conformances. Present lot is not cleared for dispatch.